

Dr. Sabina Jehan Haque

James Van Loo Independent Postdoctoral Research Fellow
University of Michigan, Department of Mathematics

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Overview

I am an applied mathematician working at the intersection of graph theory, dynamical systems, and algebraic geometry with applications in chemical reaction networks and structural identifiability. I am currently a [Van Loo Postdoctoral Research Fellow](#) jointly affiliated with the [Department of Mathematics](#) and [Center for the Study of Complex Systems](#) at the University of Michigan Ann Arbor. I completed my PhD in [Systems Biology](#) at Harvard University advised by [Jeremy Gunawardena](#). My doctoral research focused on using the linear framework, a graph-theoretic approach to continuous-time Markov processes, to characterize the interplay between network topology and system dynamics in biochemical reaction networks. My postdoctoral work has expanded this program toward structural identifiability, model reduction, and convex optimization, while developing new connections between network topology and active biological systems.

Research interests: algebraic graph theory, chemical reaction network theory, structural identifiability, real algebraic geometry, continuous-time Markov processes, non-equilibrium biophysics, dynamical systems theory, fast-slow dynamics and model reduction

Employment	University of Michigan Ann Arbor Van Loo Independent Postdoctoral Research Fellow	Jul 2024 - present
Education	Harvard University PhD in Systems Biology Advisor: Jeremy Gunawardena <i>Dissertation title: Graph-theoretic approaches to biochemical reaction networks</i>	Sep 2018 - May 2024
	Middlebury College B.A. in Mathematics (high honors), Molecular Biology & Biochemistry	Sep 2014 - May 2018

Publications

1. **Haque, S. J.**, Satriano, M., Sorea, M. & Yu, P. Y. The Disguised Toric Locus and Affine Equivalence of Reaction Networks. *SIAM J. Appl. Dyn. Syst.* **22**, 1423–1444 (2023)
2. Chavez, A., Tuttle M, Pruitt B. W., Ewen-Campen B., Chari R., Ter-Ovanesyan D., **Haque S. J.** *et al.* Comparison of Cas9 activators in multiple species. *Nat. Methods* **13**, 563–567 (2016)

In Preparation

3. **Haque S. J.**, Cetiner U., Gunawardena J. 2026. Anomalous behaviour of the Steinberg signature for detecting departure from thermodynamic equilibrium. (In preparation).
4. **Haque S. J.**, Nam K.-M., Yordanov P., Gunawardena J. 2026. A graph construction for analysing the parametric asymptotics of Markov processes (In preparation).
4. **Haque S. J.**, Eisenberg, M. 2026. Exploring the effect of parametric model reduction on the structural identifiability of linear models. (In preparation).

Fellowships and Awards

James Van Loo Independent Postdoctoral Fellowship Michigan Center for Applied and Interdisciplinary Mathematics (MCAIM), University of Michigan Ann Arbor	2024 - 2027
Junior Scientists Board Research Grant Challenge (\$10K funding) Center for Complex Particle Systems (COMPASS), University of Michigan Ann Arbor	2024 - 2025
Lynch Foundation PhD Fellowship Department of Systems Biology, Harvard University	2019 - 2021, 2023 - 2024
NSF-Simons QBio PhD Fellowship NSF-Simons Center for Mathematical & Statistical Analysis of Biology, Harvard University	2021 - 2022

Selected Talks

Graph-theoretic approaches to biochemical reaction networks and structural identifiability Applied Algebra Group, Department of Mathematics, University of Copenhagen, Copenhagen Denmark	Feb 05 2025
Graph-theoretic and algebraic geometric approaches to biochemical reaction networks Mathematics of Reaction Networks (MoRN) Seminar, virtual	Nov 11 2025
Graph-theoretic and algebraic geometric approaches to biochemical reaction networks Mathematics Seminar, Algebraic Systems Biology Group, MPI-CBG, Dresden Germany	Oct 03 2025

A graph construction for analyzing the parametric asymptotic of Markov processes 2025 SIAM Conference on Applied Algebraic Geometry, 4th Madison Workshop on Mathematics of Reaction Networks, Madison WI	Jul 12 2025
The linear framework: graph-theoretic approach to Markov processes with applications in biochemical reaction networks. Mathematics Department, University of Michigan, Ann Arbor MI	Oct 03 2024
Following the energy: graph-theoretic models of broken detailed balance with biochemical applications. National Institute for Theoretical and Mathematical Biology, Chicago IL	Jan 12 2024
Following the energy: graph-theoretic models of broken detailed balance with biochemical applications. Barcelona Collaboratorium for Modelling and Predictive Biology, UPF, Barcelona Spain	Oct 17 2023
A graph-theoretic approach to Markov processes with applications in biochemical reaction networks. Mathematics Department Colloquium, Middlebury College, Middlebury VT	Oct 03 2023
Following the energy: graph-theoretic models of broken detailed balance with biochemical applications. NSF-Simons QBio Seminar, Harvard University, Cambridge MA	Apr 25 2023
Following the energy: graph-theoretic models of broken detailed balance with biochemical applications. Systems Biology Department Seminar, Harvard University, Cambridge MA	Mar 26 2023
Graph-theoretic models of detecting broken detailed balance in molecular information processing. 2022 SIAM Annual Meeting MS 26: Trends and New Results in Deterministic Models of Biochemical Interaction Networks, Pittsburgh PA	Jul 11 2022
Graph-theoretic models of non-equilibrium conditions in molecular information processing. Systems Biology PhD Program retreat, Harvard University, Cambridge MA	May 17 2022
Investigating mathematical properties of non-equilibrium signatures in biological information processing systems. Poster at 2022 American Physical Society Annual March Meeting, Chicago IL	Mar 16 2022
Graph-theoretic models of non-equilibrium conditions in molecular information processing. NSF-Simons QBio Seminar, Harvard University, Cambridge MA	Nov 10 2021

University of Michigan Ann Arbor

Instructor of record:

MATH 404: Intermediate Differential Equations (Nonlinear Dynamics & Chaos)	Winter 2025
MATH 404: Intermediate Differential Equations (Nonlinear Dynamics & Chaos)	Fall 2025
MATH 404: Intermediate Differential Equations (Nonlinear Dynamics & Chaos)	Fall 2026

Undergraduate students mentored:

- Joe Bogdan (University of Michigan, Independent Research Study, Summer 2025)
- Daniel Le (Colby College, Math REU, Summer 2026)
- George Simmons (University of Michigan, Math REU, Summer 2026)

Harvard University

AM 50: Introduction to Applied Mathematics	Spring 2020
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Conferences and workshops attended

*14th European Conference on Mathematical & Theoretical Biology University of Graz, Austria	Jul 2026
*Chemical Reaction Networks Workshop in Hawai'i 2026 University of Hawai'i at Mānoa, Honolulu, HI	May 2026
*4th Madison Workshop on Mathematics of Reaction Networks University of Wisconsin-Madison, Madison WI	Jul 2025
2025 SIAM Conference on Applied Algebraic Geometry Meeting University of Wisconsin-Madison, Madison WI	Jul 2025
Simons-NSF MathBioSys Annual Meeting 2023 Simons Foundation, New York NY	Apr 2023
*2022 SIAM Annual Meeting David L. Lawrence Convention Center, Pittsburgh PA	Jul 2022
*APS Annual March Meeting 2022 McCormick Place - West Building, Chicago IL	Mar 2022
Quantitative Approaches in Biology (virtual) Northwestern University NSF-Simons Center	Nov 2020
Mathematical Models in Biology: From Information Theory to Thermodynamics (virtual) Banff International Research Station (BIRS)	Jul 2020
Workshop on Dynamics, Randomness, and Control in Molecular and Cellular Networks	Nov 2019

Center for Mathematical Sciences and Applications (CMSA), Harvard University, Cambridge MA

Outreach

MCAIM Colloquium organizer University of Michigan, Ann Arbor MI	2026-present
American Women in Mathematics peer mentor University of Michigan, Ann Arbor MI	2024-present
Guest speaker for summer class on Physics of Biological Function Wentworth Institute of Technology, Boston MA	Jun 2023
Cambridge Science Festival volunteer Cambridge, MA	Oct 2022
Graduate research assistant in Quantitative Biology grant preparation NSF-Simons Center for Quantitative Biology, Harvard University, Cambridge MA	May-Jul 2022
Diversity, equity, and inclusion contributor, recruiter, and mentor Systems Biology Department/PhD Program, Harvard University, Cambridge MA	2020-2024
“What is Systems Biology?” outreach event creator/coordinator Cambridge Science Festival, Cambridge MA	Apr 2019

Science communication

Cellular mathematics: how does math enhance our understanding of life at the molecular level? https://sabinahaque.substack.com/	2022-present
Graph Theory 101 Science in the News Special Edition: Networks, Harvard University, Cambridge MA https://sitn.hms.harvard.edu/flash/2021/graph-theory-101/	Jun-Aug 2021

References

References made available upon request.